

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 1

Naresh Manwani

Machine Learning Lab, IIIT-H

July 31st, 2026



Course Syllabus

- ① Sample Spaces, Event Spaces, Counting, Uniform Probability, Axioms of Probability, Continuous Probability Spaces, Conditional Probability, Bayes Rule, Independence.
- ② Random Variables, Probability Distributions of Random Variables, Discrete and Continuous Random Variables, Functions of Random Variables.
- ③ Multiple Random Variables, Independent Random Variables, Functions of Several Random Variables, Expectation, Variance, Covariance, Correlation, Conditional Expectation, Order Statistics and Their Distributions.
- ④ Some Discrete Distributions, Bernoulli, Binomial, Geometric, Poisson, Multinomial. Some Continuous Distributions, Uniform, Gaussian, Exponential, Gamma, Beta. Bivariate and Multivariate Normal Distributions.
- ⑤ Tail Bounds (Markov, Chebyshev, Chernoff), Weak Law of Large Numbers, Strong Law of Large Numbers, Central Limit Theorem.
- ⑥ Maximum A posteriori Estimation, Maximum Likelihood Estimation, Confidence Intervals.

- 1 Introduction to Probability, 2nd Edition by Dimitri P. Bertsekas and John N. Tsitsiklis.
- 2 Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross.
- 3 An Introduction to Probability Theory and Its Applications, Volume 1 by William Feller.
- 4 Probability, Random Variables and Stochastic Processes by Athanasios Papoulis and S. Unnikrishna Pillai. 4th Edition.

Course Evaluation

- ① Assignments (3 Hand written) : 30% (10% weightage for each assignment. After every 2 weeks, one assignment will be released.)
- ② Quiz 1: 20%
- ③ Mid : 35%
- ④ Surprise Quizzes in Tutorials ($3 \times 5\%$): 15%
 - Will be announced 1 day before the tutorial about the quiz.
 - If you miss any of the quiz, it will not be repeated. No appeal for re-exam will be accepted.

Tutorial Timing : 2.00-3.00 PM (Every Monday)

- 1 Ayush Mahesh Sheta (ayush.sheta@students.iiit.ac.in)
- 2 Sravan Kotta (sravn.kotta@research.iiit.ac.in)
- 3 Vadla Shiva Kishan (vadlashiva.kishan@students.iiit.ac.in)
- 4 Kushal Mukherjee (kushal.mukherjee@students.iiit.ac.in)
- 5 Varun Gupta (varun.gup@students.iiit.ac.in)

What is Probability

Read the sentence.

“If I flip this coin, the probability of getting heads is 0.5”

What does this mean?

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- If I were to toss the coin 10 times, roughly 5 times I will see a head.
- A probability of 1 means it is certain, a probability of 0 means it is impossible.
- In general, you do an experiment many times, and you count how many times a particular event occurs. The proportion roughly gives you the probability of that particular event.

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- Experiment 2: You throw two dice
 - Event 1: the sum of the rolls is six
 - Event 2: you get two odd faces

Sample Space

Definition

Sample Space: The sample space Ω is the set of all possible outcomes of an experiment. The different elements of a sample space must be mutually exclusive and collectively exhaustive.

Examples of Experiments and their Sample Spaces:

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- Experiment: Two cards are drawn from an ordinary deck of cards (i) with replacement and (ii) without replacement.
 - **With replacement:** Ω contains 52×52 possible pairs of cards.

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- Experiment: Two cards are drawn from an ordinary deck of cards (i) with replacement and (ii) without replacement.
 - **With replacement:** Ω contains 52×52 possible pairs of cards.
 - **Without replacement:** Ω contains 52×51 possible pairs of cards. Thus, it does not contain pairs in which both cards are exactly same.

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 - **Event:** $E = \{HH, HT\}$, then E is the event that head appears on the first coin.

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- **Experiment:** A coin is tossed until first head appears ($\Omega = \{H, TH, TTH, TTTH, TTTTH, \dots\}$).
 - **Event:** $E = \{H, TH, TTH\}$, the E is the event that in at most third toss you see the first head.

Sets and Sample Spaces

We need to introduce some mathematical concepts to define probability more concretely:

Sets

A set is a collection of objects, which are called elements.

- If an element x is in a set S , we write $x \in S$.
- If a set contains no elements, we call it the empty set, ϕ .
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Examples of Sets

- The natural numbers are a set, where the elements are individual numbers.
- This class is the set, where the elements are the professor, the TA and the students.

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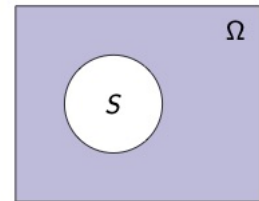
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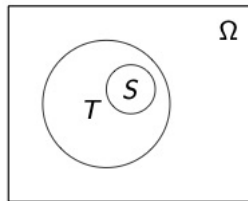
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 - the set of all subsets of natural numbers, aka the power set

Operations on Sets

- Let the **universal set** Ω be the set of all objects we might possibly be interested in.



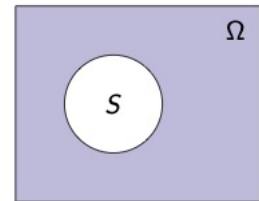
S^c is the shaded region



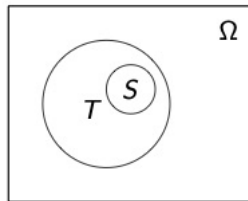
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- Let the **universal set** Ω be the set of all objects we might possibly be interested in.
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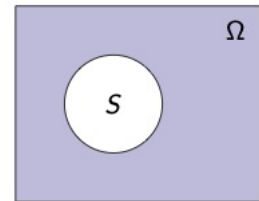
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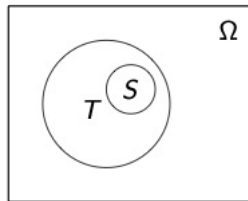
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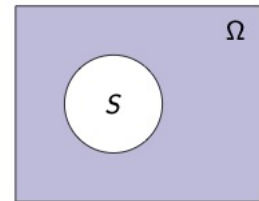
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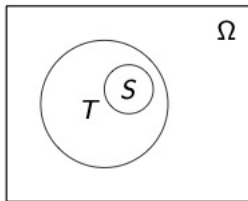
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- $S \subseteq T$ and $T \subseteq S$ if and only if $S = T$.



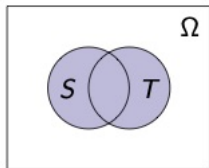
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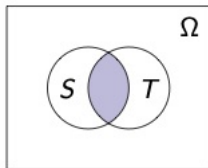
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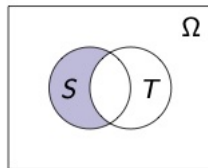
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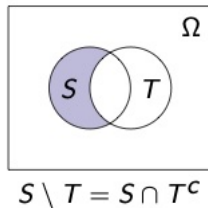
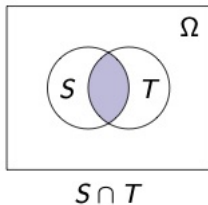
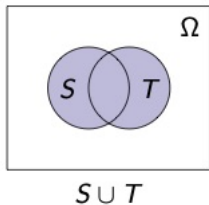
$$S \cap T$$



$$S \setminus T = S \cap T^c$$

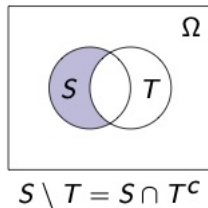
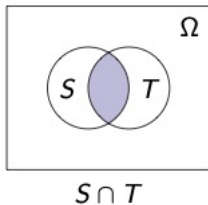
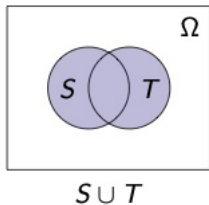
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- The **difference**, $S \setminus T$, of two sets S and T is the set of elements that are in S , but not in T : $S \setminus T = \{x \mid x \in S \text{ and } x \notin T\}$.



- We can extend the notions of union and intersection to multiple (even infinitely many!) sets:

$$\bigcup_{i=1}^n S_i = S_1 \cup S_2 \cup \dots \cup S_n = \{x \mid x \in S_i \text{ for some } 1 \leq i \leq n\}$$

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- We say a collection of sets are **disjoint** if no two sets have any common elements.
- If a collection of disjoint sets have union S , we call them a partition of S .

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Axioms of Probability:

- **Nonnegativity:** $P(A) \geq 0$, for every event A .
- **Additivity:** Let $\{A_j\}_{j=1}^{\infty}$ be a sequence of disjoint sets such that $A_j \in \Omega$, $\forall j = 1 \dots \infty$. Thus $A_i \cap A_j = \phi$, $\forall i \neq j$. Then,

$$P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$$

Probability of an Event

The probability law assigns to an event E a non-negative number $P(E)$ which encodes our belief/knowledge about the “likelihood” of the event E .

Axioms of Probability:

- **Nonnegativity:** $P(A) \geq 0$, for every event A .
- **Additivity:** Let $\{A_j\}_{j=1}^{\infty}$ be a sequence of disjoint sets such that $A_j \in \Omega$, $\forall j = 1 \dots \infty$. Thus $A_i \cap A_j = \phi$, $\forall i \neq j$. Then,

$$P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$$

- **Normalization:** The probability of the entire sample space Ω is equal to 1, i.e. $P(\Omega) = 1$.

You tossed two fair dice together. What is the probability of the event $E = \{\text{sum of the rolls} = 6\}$?

- The sample space is $\Omega = \{(i, j) \mid 1 \leq i, j \leq 6\}$. There are a total of 36 outcomes.
- Since the dice are fair, each outcome is equally likely. This means every outcome has probability $1/36$.
- We get a sum of 6 if we get $E = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$.
- Using the additivity law we have $P(E) = 5/36$.

This is an example of an uniform distribution, where all outcomes are equally likely.

- If $A \subseteq B$, then $P(A) \leq P(B)$.

Properties of Probability Laws

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- $P(A \cup B \cup C) =$
 $P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

Theorem

Let $A_1, A_2, \dots, A_n \in \Omega$. Then

$$\begin{aligned} P(\cup_{i=1}^n A_i) &= \sum_{i=1}^n P(A_i) - \sum_{i_1=1}^n \sum_{i_2 > i_1} P(A_{i_1} \cap A_{i_2}) \\ &\quad + \sum_{i_1=1}^n \sum_{i_2 > i_1} \sum_{i_3 > i_2} P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) + \dots \\ &\quad + (-1)^{n+1} P(\cap_{i=1}^n A_i) \end{aligned}$$

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How do you prove it? [Hint: How about Induction!]